VASANTHA DEVI KANDASAMY

AI Engineer · Full-Stack Developer · NLP & Data Systems

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SUMMARY

Full-stack engineer specializing in applied AI and natural-language interfaces to data. I build LLM systems that turn plain-English questions into grounded, verified answers over real databases — natural-language-to-SQL, retrieval-augmented generation (RAG with pgvector), and tool-using agents — engineered so results are grounded in structured data, not hallucinated. 3+ years of full-stack development (React/Node) plus recent data-analytics training (IBM Data Analyst, Azure AI-900). Shipped four containerized, eval-tested AI applications, including a natural-language-to-SQL analytics copilot (100% execution accuracy). Comfortable owning a problem from the first customer conversation to a deployed, observable solution.

TECHNICAL SKILLS

AI & LLM: Claude (Anthropic) API, LLM agents & tool use, RAG / semantic search (pgvector), text-to-SQL (NL→SQL), Natural Language Understanding (NLU), grounded / deterministic AI, Model Context Protocol (MCP / FastMCP), prompt engineering, evaluation suites, LLM observability (Langfuse), token streaming (SSE)

Backend & Data: Python (FastAPI), Node.js (Express), SQL, PostgreSQL + pgvector, MongoDB, REST APIs, JWT/OAuth; Pandas, NumPy, data analysis, dashboarding

Frontend & DevOps: JavaScript, React, Redux, Tailwind CSS; Docker, Kubernetes (GKE), Google Cloud (Cloud Run, Cloud SQL), CI/CD, Git, Playwright

AI PROJECTS

Analytics Copilot — Natural-language-to-SQL over a database

Python · FastAPI · Claude API · PostgreSQL + pgvector · React · Docker / GCP · github.com/vasanthadevi/analytics-copilot

- Built a copilot that answers plain-English business questions: retrieves the relevant tables via semantic search (RAG with pgvector), generates read-only SQL, self-corrects on query errors, and returns a chart, a table, and the exact SQL it ran.
- Guaranteed safety with a dedicated read-only database role (SELECT-only, statement timeout) so model-generated SQL can never modify data; authored a 40-question execution-accuracy eval suite and improved accuracy from 97.5% to 100% with targeted prompt changes.

AI Support Assistant — Streaming, grounded customer-support chat

Node.js · Express · Claude API · PostgreSQL · React · Docker · github.com/vasanthadevi/ai-support-assistant

- Built a real-time assistant that streams answers token-by-token (SSE) and calls tools to search a knowledge base and look up live order data mid-conversation — citing its sources, refusing to guess, and tracking per-message token cost.

Nimbus Operations Copilot — Agentic operations assistant (capstone)

Python · FastMCP (MCP) · Claude API · FastAPI · React · PostgreSQL · Langfuse · Docker / GCP · github.com/vasanthadevi/ops-copilot

- Built a multi-step agent that plans across tools exposed through a custom MCP server; enforced human-in-the-loop approval before any data-changing action, with Langfuse tracing and a 10-task eval suite (10/10 correctness, zero writes executed without approval).

Singh's Auto Lab AI Service Advisor — Agentic web assistant for a real Calgary business

Python · FastAPI · 3× FastMCP servers · Claude API · React · Google Calendar · Twilio · Docker · github.com/vasanthadevi/singhs-auto-lab-advisor

- Triages a customer's problem, returns a grounded estimate from a structured pricing knowledge base (not the model), books into a real Google Calendar, and recognizes returning customers via a phone-keyed CRM; capabilities split into repointable MCP servers, plus vision photo-diagnosis and an owner dashboard.

EXPERIENCE

Front-End Developer (Volunteer) — CMDSpace, Calgary *Jan – Jun 2024*

- Built the front end for an AI meeting-organizer app (React, Node, Express, MySQL, Playwright); dockerized the app and ran containerized tests for reliable deployment.

Freelance Web Developer — Upwork (Haydes Technologies) *2013 – 2015*

- Developed, maintained, and supported web applications; built secure login/authentication; delivered front-end training to associates.

Software Developer — Mascot eSoft / ISG Novasoft, India *2006 – 2007*

- Built web applications (SQL Server, VB.NET/ASP.NET, JavaScript/jQuery), including an RFID-based access-control and reporting system.

EDUCATION & CERTIFICATIONS

Master of Computer Applications (MCA), Bharathiar University, India · **Bachelor of Computer Applications (BCA)**

IBM Full-Stack Software Developer · IBM Data Analyst (Coursera, 2024) · NPower Canada Junior Data Analyst (2024) · Microsoft Azure AI-900 (2024)